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MIL-M-28787/322B
12 October 1994
SUPERSEDING
MIL-M-28787/322A(EC)
30 October 1984

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, D/A, KEY CODE DBK

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

*The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic D/A converter module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

*(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 22.95 failures per million operating hours for class I or 83.64 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

* 3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

* 3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DBK.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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* TABLE I Module electrical requirements

	Test cond	Parameter	Ref test table	Test type 1/ 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
							0°C ± 5°C		60°C ± 5°C		55°C ± 5°C		85°C ± 5°C		
					Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	A	Stress	II	Q	-100	100	100	100	100	100	-100	100	100	100	mV
*	B	Short circuit current (Vout)	III	Q	50	-5	50	5	-50	5	-50	5	50	-5	mA
	C1 C5	Isolation													
		Initial	IV	X	10	INF	10	INF	10	INF	10	INF	10	INF	MOhm
		End-of-life	IV	X	1	INF	1	INF	1	INF	1	INF	1	INF	MOhm
*	D1 D2	Power supply current	V	X	1	55	1	55	1	55	1	55	1	55	mA
*	D3 D4	Power supply current	V	X	1	40	1	40	1	40	1	40	1	40	mA
	E1 E8	Continuity	VI	X	0	1	0	1	0	1	0	1	0	1	Ohm
	E9	Resistance													
		Initial	VI	X	4 46	4 94	4 32	5 08	4 32	5 08	3 99	5 41	4 28	5 12	kOhm
		End of life	VI	X	4 18	5 22	4 04	5 36	3 99	5 41	3 71	5 69	3 90	5 50	kOhm
	F1 F2	Capcitance, power supply	VII	X	0 75	1 25	0 75	1 25	0 75	1 25	0 75	1 25	0 75	1 25	μF
*	G1 G12	Digital input current, high level	VIII	X	10	40	10	40	-10	40	-10	40	10	40	μA
	H1 H12	Digital input current low level	IX	X	-1 6	0 0	1 6	0 0	-1 6	0 0	-1 6	0 0	-1 6	0 0	mA
		Bipolar accuracy,													
	J1 J3	± 2 5 V range	X	X	10	10	15	15	-15	15	-20	20	-20	20	LSB
	J2	± 2 5 V range	X	X	-20	20	30	30	30	30	30	30	-30	30	mV
	J4 J6	± 5 V range	X	X	-10	10	15	15	-15	15	20	20	-20	20	LSB
	J5	± 5 V range	X	X	-30	30	-40	40	40	40	40	40	40	40	mV
	J7 J9	± 10 V range	X	X	10	10	15	15	15	15	20	20	20	20	LSB
	J8	± 10 V range	X	X	-50	50	-60	60	60	60	-60	60	60	60	mV
	K1 K32	Unipolar accuracy	XI	X	1	1	4	4	4	4	-10	10	-10	10	LSB
*	L1 L15	Differential linearity error	XII	X	0	4 88	0	4 88	0	4 88	0	4 88	0	4 88	mV
*	M1 M30	Linearity error	XIII	X	-2 44	2 44	2 44	2 44	2 44	2 44	-2 44	2 44	-2 44	2 44	mV
*	N1 N4	Power supply sensitivity	XIV	S	-6 1	6 1	6 1	6 1	6 1	6 1	-6 1	6 1	6 1	6 1	mV

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• TABLE I Module electrical requirements - Continued

Test cond	Parameter	Ref test table	Test type 1/ - 3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	Settling time to													
P1 P3	± 1/2 LSB (FS chng)	XV	Q	0 0	5 0	0 0	5 0	0 0	5 0	0 0	5 0	0 0	5 0	µS
P2 P4	± 2 LSB (FS chng)	XV	Q	0 0	3 0	0 0	3 0	0 0	3 0	0 0	3 0	0 0	3 0	µS
P5,P7	± 1/2 LSB (FS/2 chng)	XV	Q	0 0	4 5	0 0	4 5	0 0	4 5	0 0	4 5	0 0	4 5	µS
P6 P8	± 2 LSB (FS/2 chng)	XV	Q	0 0	2 5	0 0	2 5	0 0	2 5	0 0	2 5	0 0	2 5	µS
Q	Latch	Fig 4	Q	-100	100	-100	100	-100	100	-100	100	-100	100	mV

1/ "X" parameters shall be 100 percent tested during the module acceptance test

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life

5/ Abbreviations used in table I are defined in 6 1

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/322-1 for class I and M28787/322-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

* 3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1.

3.8 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

* 3.10 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

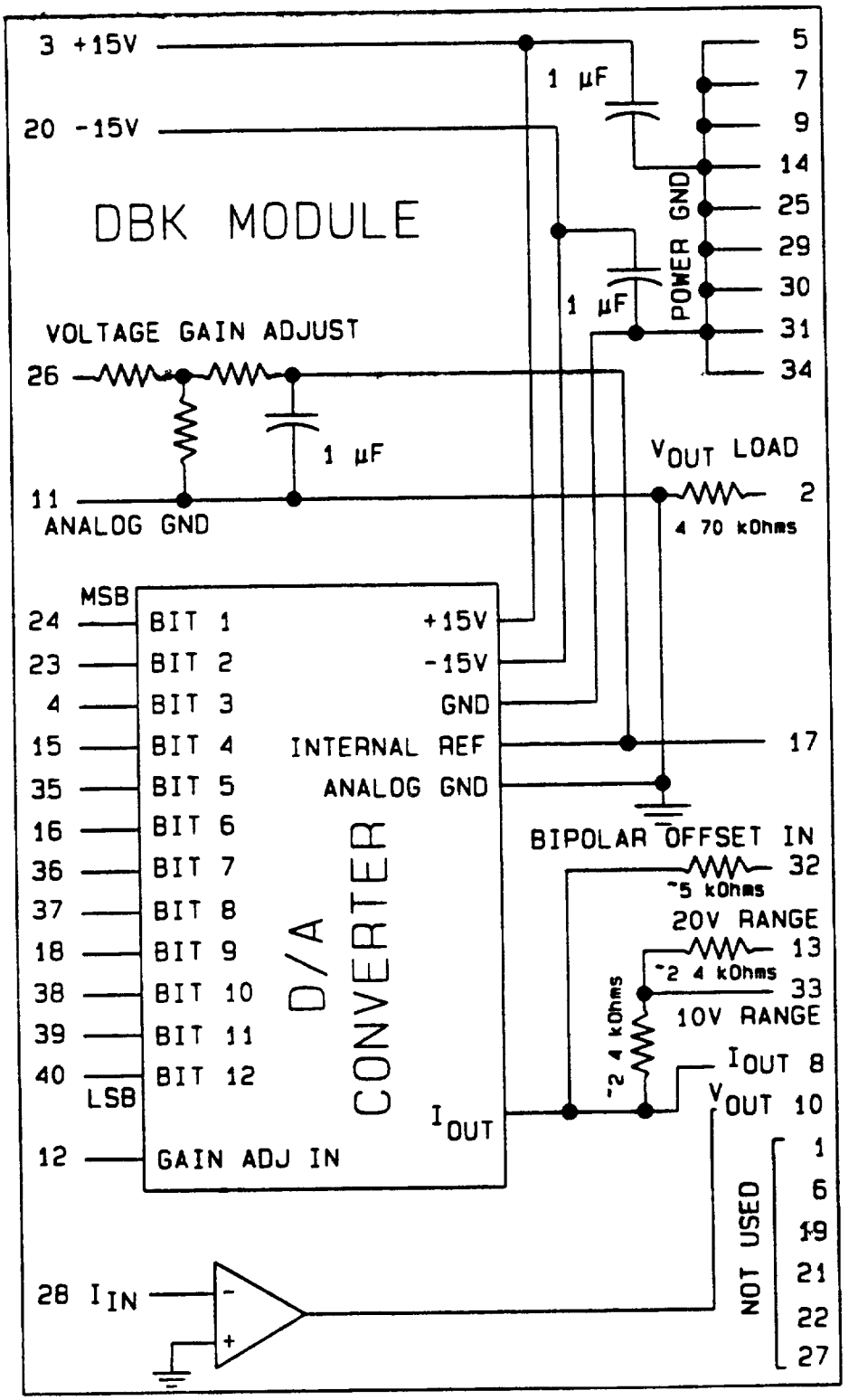
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

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* FIGURE 1. DBK functional block diagram.

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* 4.4.2 Order of testing. Test conditions A through N shall be performed in any order with the following exceptions. Test conditions A, B and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B or C are part of a sequence of tests, the order of testing shall be as follows: A, B and C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

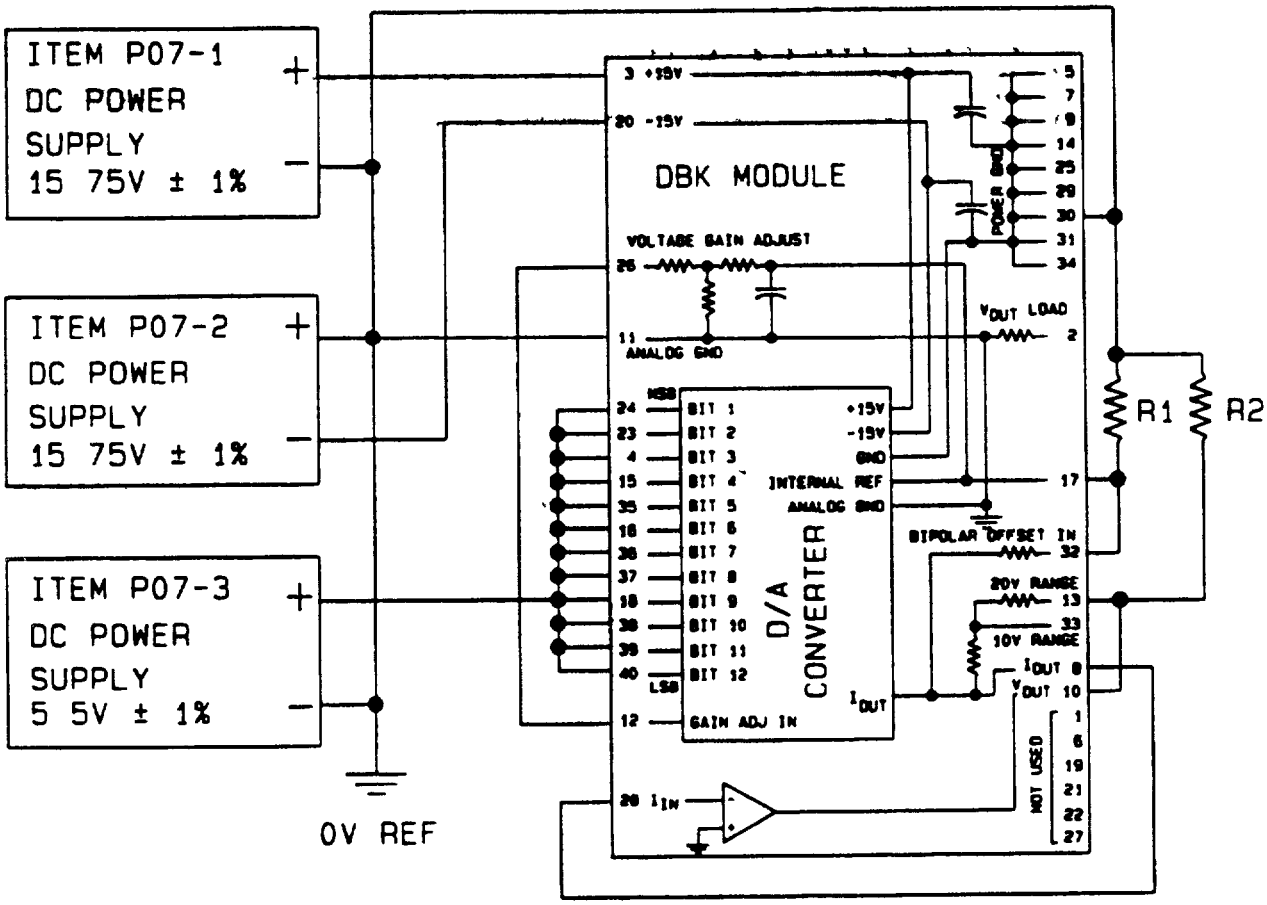
* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through N in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through N shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

* 4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test and prior to disconnecting the power, electrical test Q shall be performed in accordance with figure 4 and while still at the maximum class temperature, electrical tests D through N shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests D through N shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

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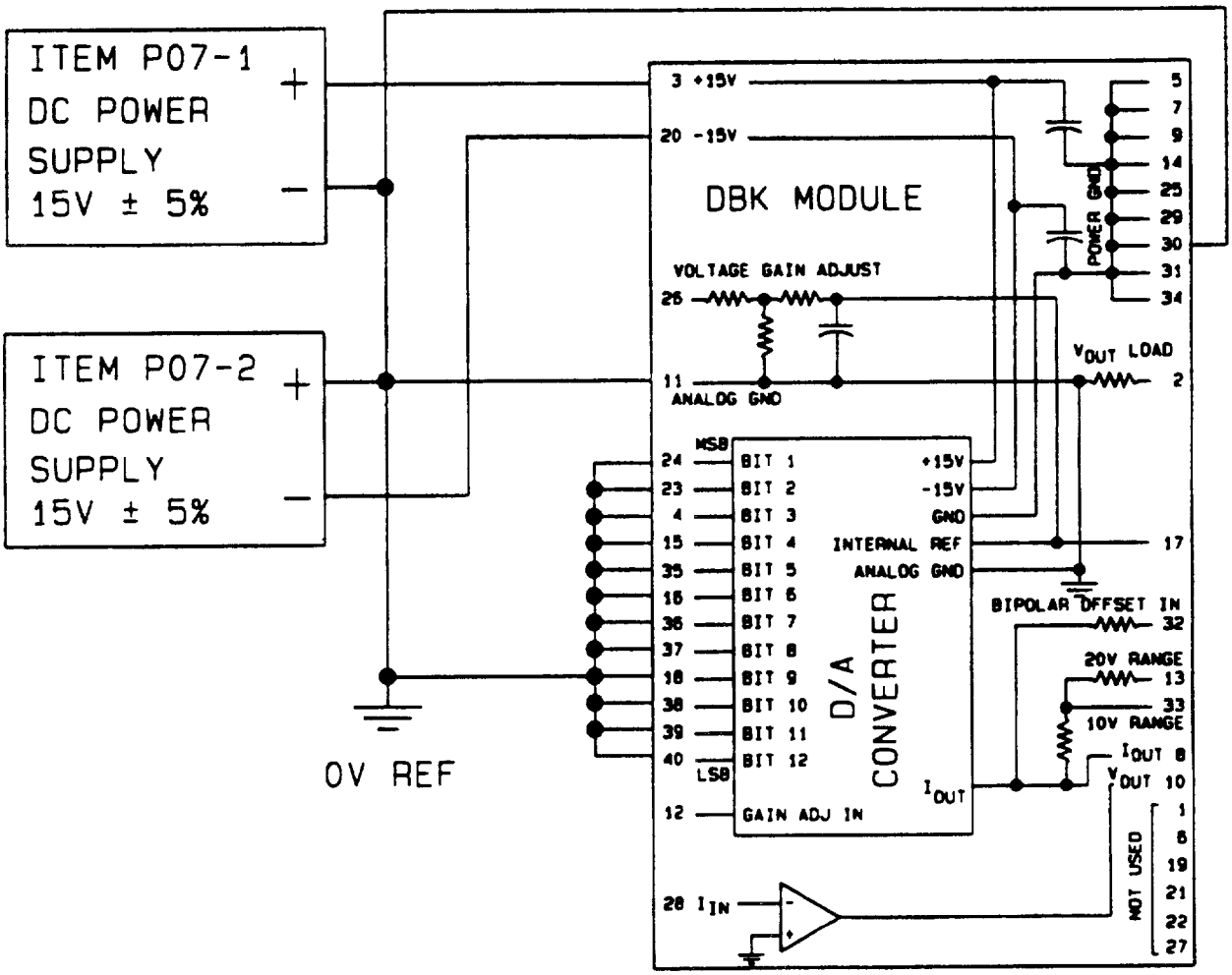


NOTES:

1. R1 = 10 kOhms ± 1 percent, 1/8 W.
2. R2 = 2 kOhms ± 1 percent, 1/8 W.

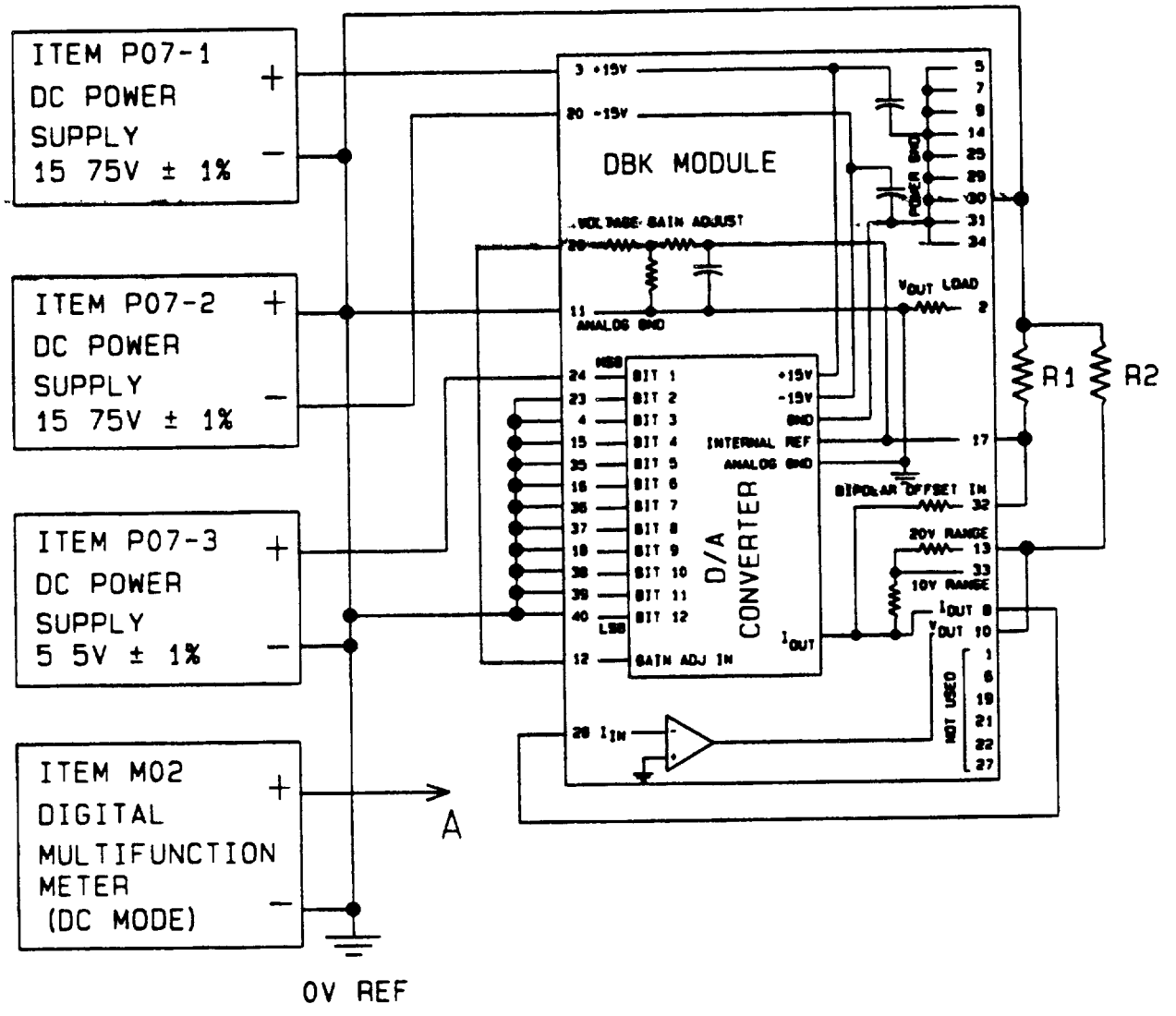
* FIGURE 2. Operating and component temperature test circuit.

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* FIGURE 3. Vibration test circuit.

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NOTES:

1. R1 = 10 kOhms ± 1 percent, 1/8 W.
2. R2 = 2 kOhms ± 1 percent, 1/8 W.
3. Following the life test period and prior to disconnecting power, connect A to pin 10. The voltage shall meet the limits specified in table I for test condition Q.
4. Test condition Q verifies that the amplifier does not latchup.

* FIGURE 4. Life test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

* 6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

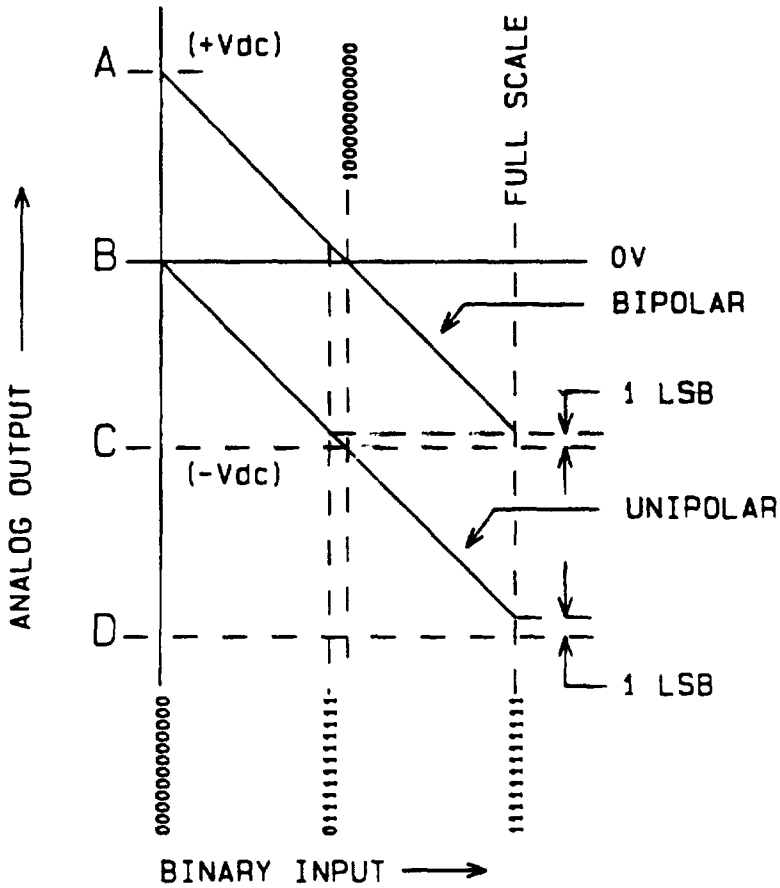
6.3 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits in table I.

	Class I	Class II	Unit
Maximum power dissipation	1.4	1.4	W
Input voltage			
Low level (maximum)	0.8	0.8	V
High level (minimum)	2	2	V
Input current			
Low level (VIL = 0.8 V)	-1.6	-1.6	mA
High level (VIH = 2.0 V)	40	40	µA
Unipolar accuracy error (0 - 10 V range)	± 4 (± 1/2 in 10 bits)	± 10 (< ± 1/2 in 8 bits)	LSB
Bipolar accuracy error	± 15	± 20	LSB

* 6.4 Module function. This module contains a 12 bit digital-to-analog converter, an operational amplifier, and a resistor network for programming signal ranges. The output voltage range is -10 to 0 V. Connecting pin 17 to pin 32 offsets the voltage by 5 V.

* 6.4.1 Input to output relationship. The voltage input to output relationship is shown on figure 5. Voltage gain correction is available by connecting pin 12 to pin 26.

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Voltage range	A	B	C	D	Unit	Connect pins
-2.5 to 2.5	2.5	0.0	-2.5	--	V	8 to 13 to 28, 10 to 33,12 to 26, 17 to 32
-5.0 to 5.0	5.0	0.0	-5.0	--	V	8 to 28, 10 to 33, 12 to 26,17 to 32
-10.0 to 10.0	10.0	0.0	-10.0	--	V	8 to 28, 10 to 13, 12 to 26,17 to 32
-5.0 to 0.0	--	0.0	-2.5	-5.0	V	8 to 13 to 28, 10 to 33,12 to 26,
-10.0 to 0.0	--	0.0	-5.0	-10.0	V	8 to 28,10 to 33, 12 to 26

* FIGURE 5. Input to output relationship.

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* 6.4.2 Previous configuration. Modules produced to this specification through and including revision A Amendment 1 used a potentiometer to adjust current gain. The current gain function of the module is not tested by this specification; therefore, the potentiometer is no longer required. However, for installations which ground all unused pins, care should be taken not to ground pin 27 such that earlier revisions of the module can be used.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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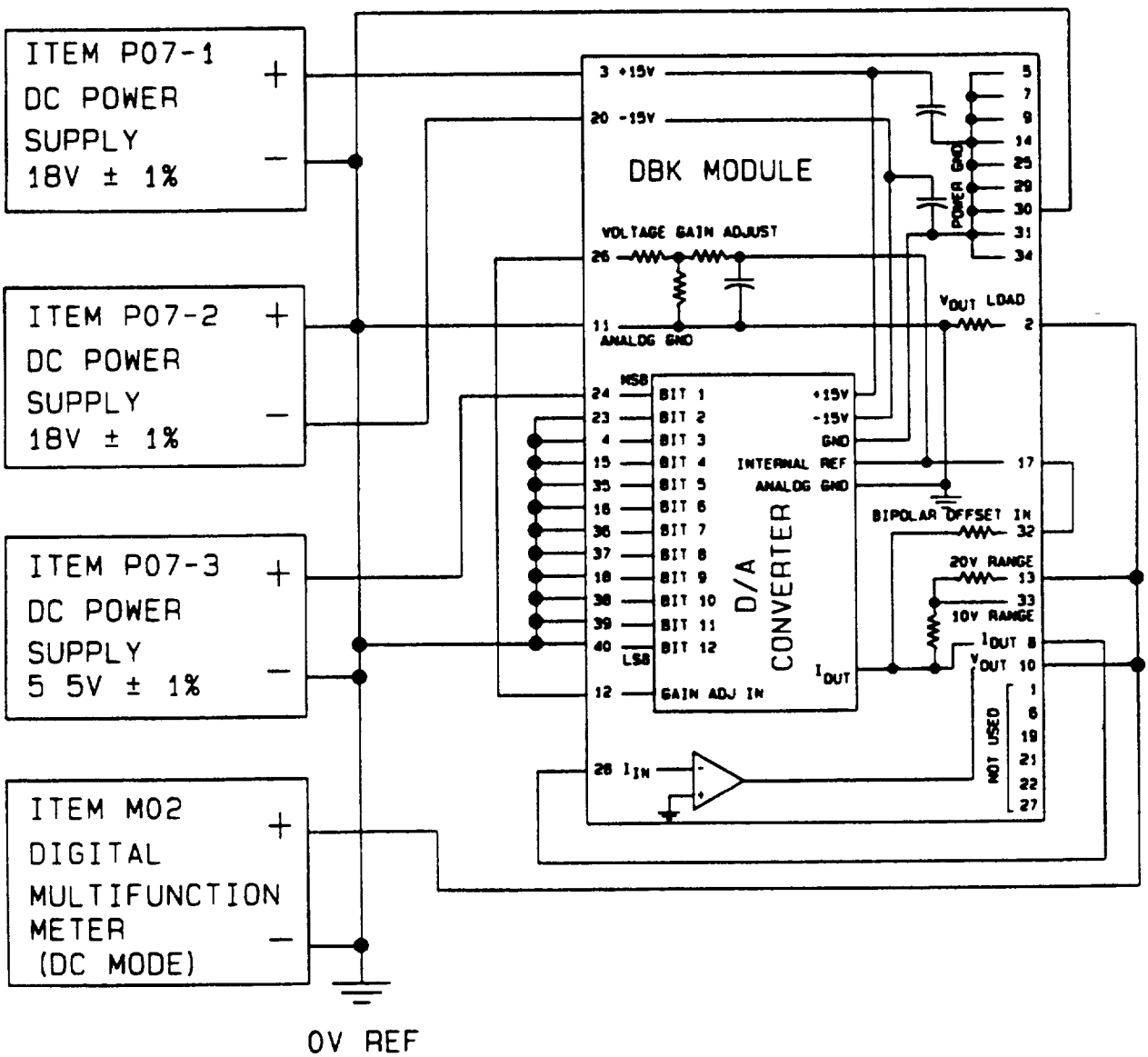
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TABLE II. Stress test.

Test cond	Parameter	Initial limits 25°C ± 5°C ambient		Unit	Test figure
		Min	Max		
A	Stress	-100	100	mV	6

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NOTE:

Wait one minute after power is applied before taking reading.

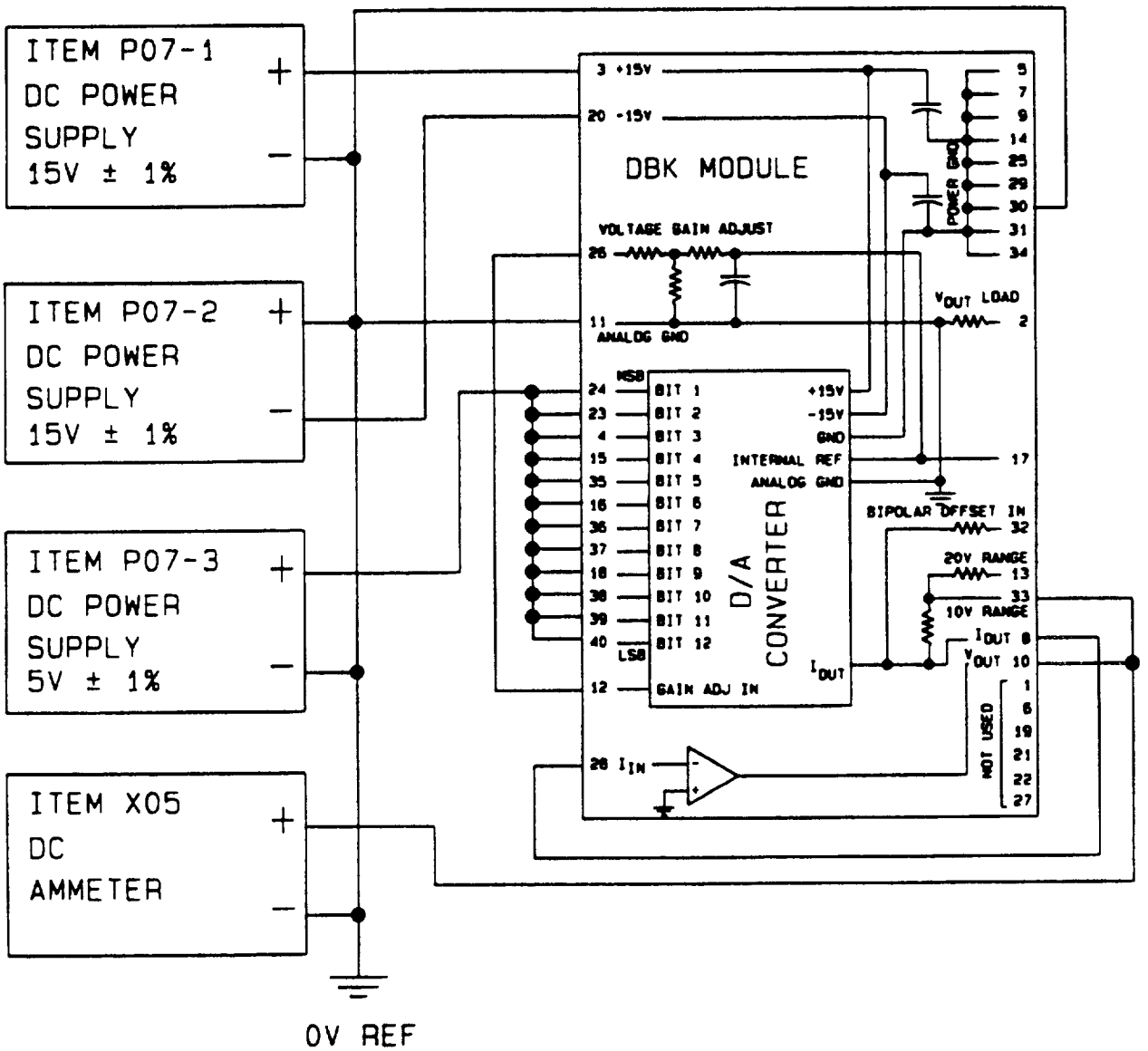
* FIGURE 6. Stress test circuit for table II.

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* TABLE III. Short circuit current (Vout) test.

Test cond	Parameter	Initial limits 25°C ± 5°C ambient		Unit	Test figure
		Min	Max		
* B	Short circuit current (Vout)	-50	-5	mA	7

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NOTE:

Wait one minute after power is applied before taking reading.

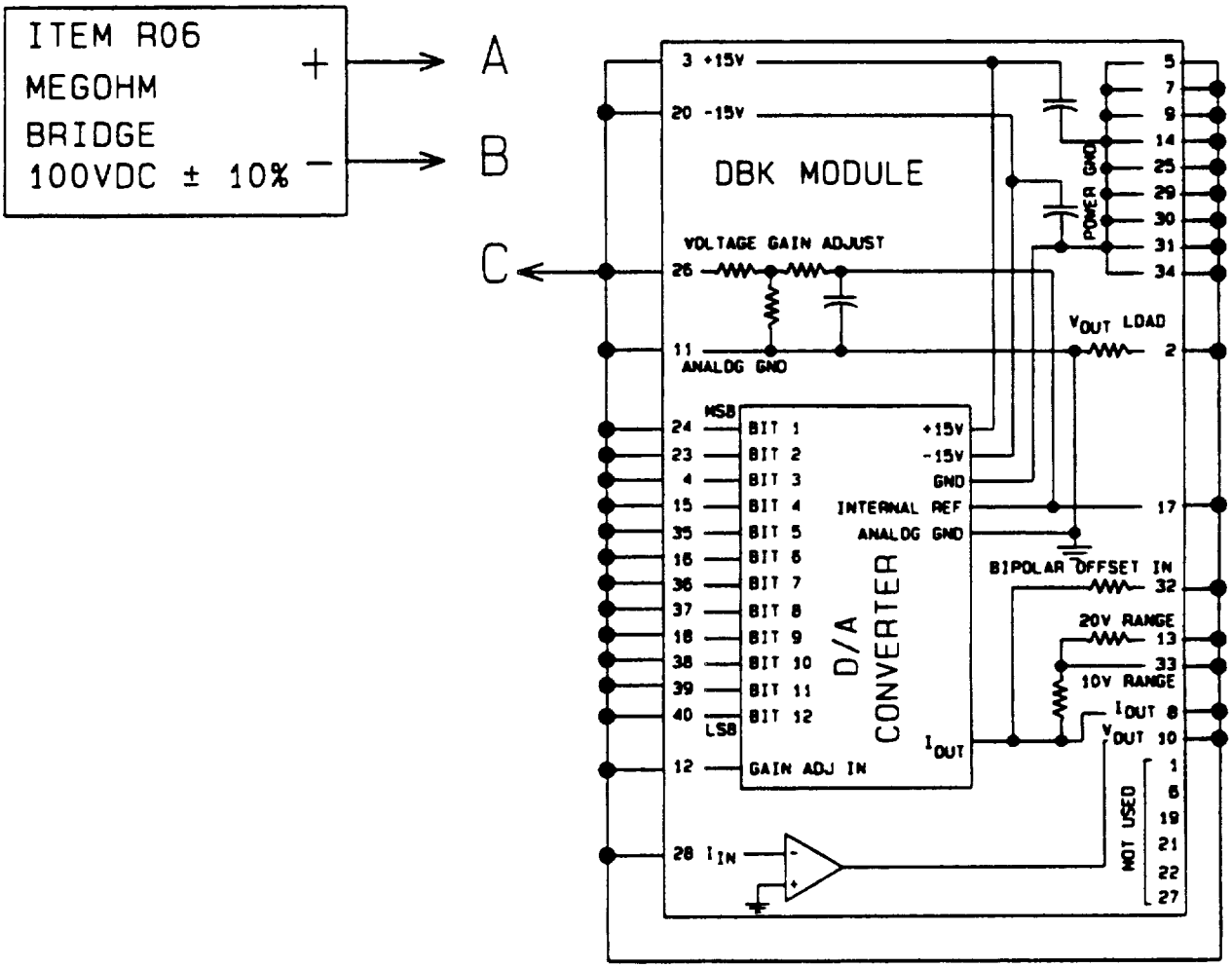
* FIGURE 7. Short circuit current test circuit for table III.

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TABLE IV. Isolation test

Test cond	Parameter	Connect		Initial limits 25°C, ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Used to unused pins	C	1, 6, 19, 21, 22	10	INF	MOhm	8
C2	Unused to unused pins	1	6, 19, 21, 22				
C3		6	19, 21, 22				
C4		19	21, 22				
C5	Unused to unused pins	21	22	10	INF	MOhm	8

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

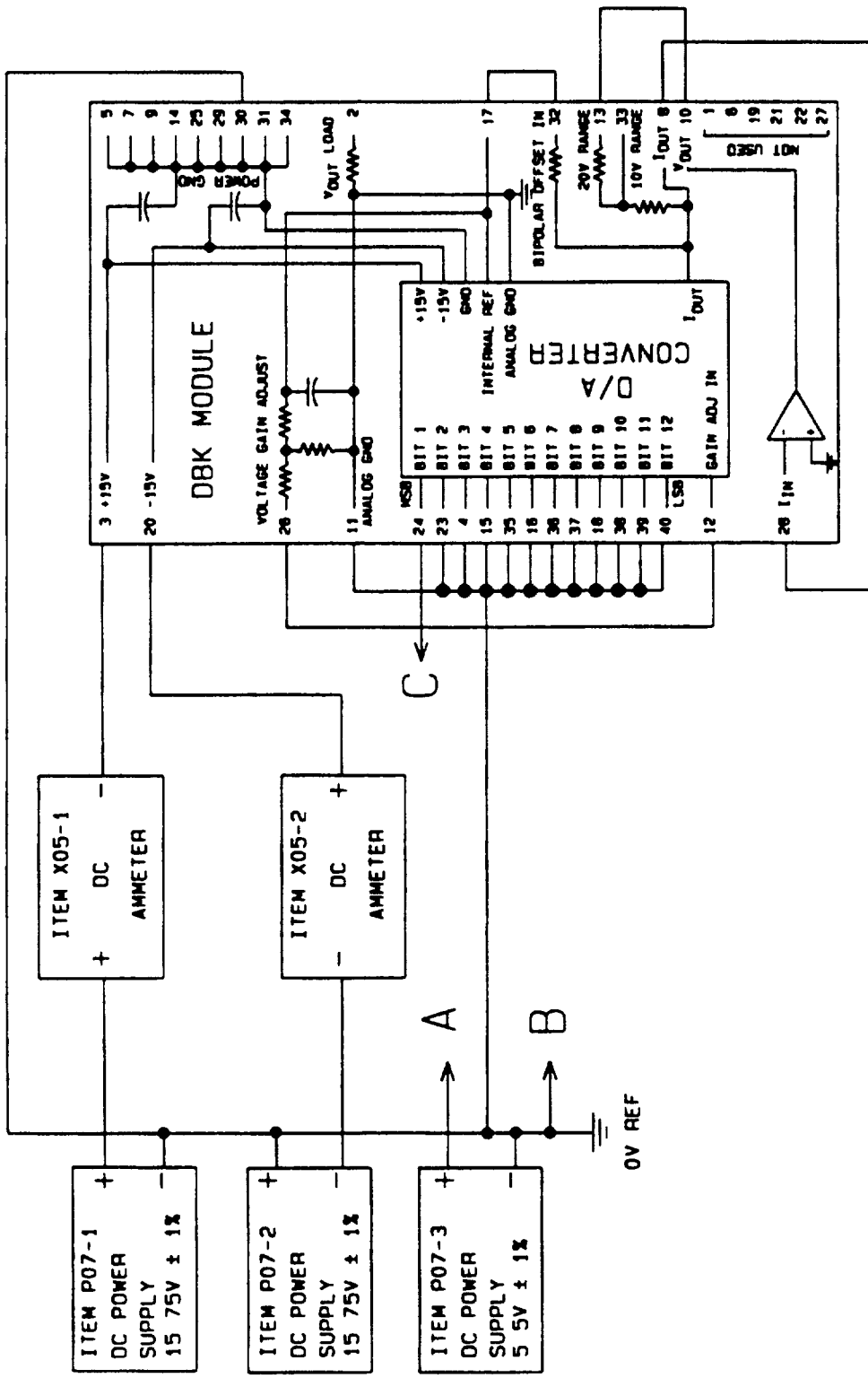
* FIGURE 8. Isolation test circuit for table IV.

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* TABLE V. Power supply current test.

	Test cond	Parameter	Connect C to point	Initial limits 25°C ± 5°C ambient		Unit	Test figure
				Min	Max		
*	D1	15 V Power supply current (all bits low)	B	1	55	mA	9
*	D2	15 V Power supply current (all bits low except MSB)	A		55		
	D3	-15 V Power supply current (all bits low)	B		40		
	D4	-15 V Power supply current (all bits low except MSB)	A	1	40	mA	9

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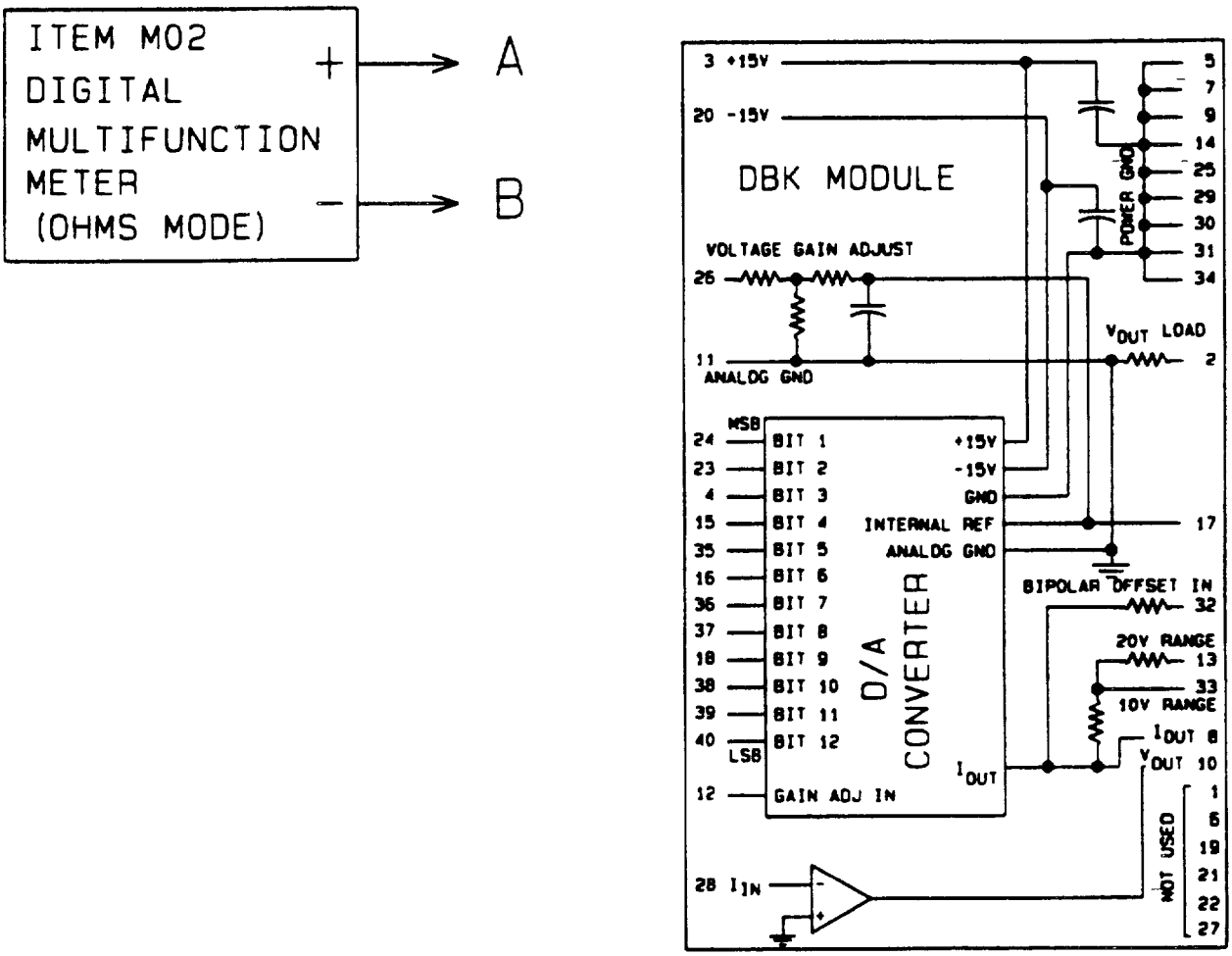
* FIGURE 9. Power supply current test circuit for table V.

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TABLE VI. Continuity and resistance test.

Test cond	Parameter	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Continuity	5	7	0	1	Ohm	10
E2			9				
E3			14				
E4			25				
E5			29				
E6			30				
E7			31				
E8	Continuity	5	34	0	1	Ohm	
E9	Resistance	2	11	4.46	4.94	kOhm	10

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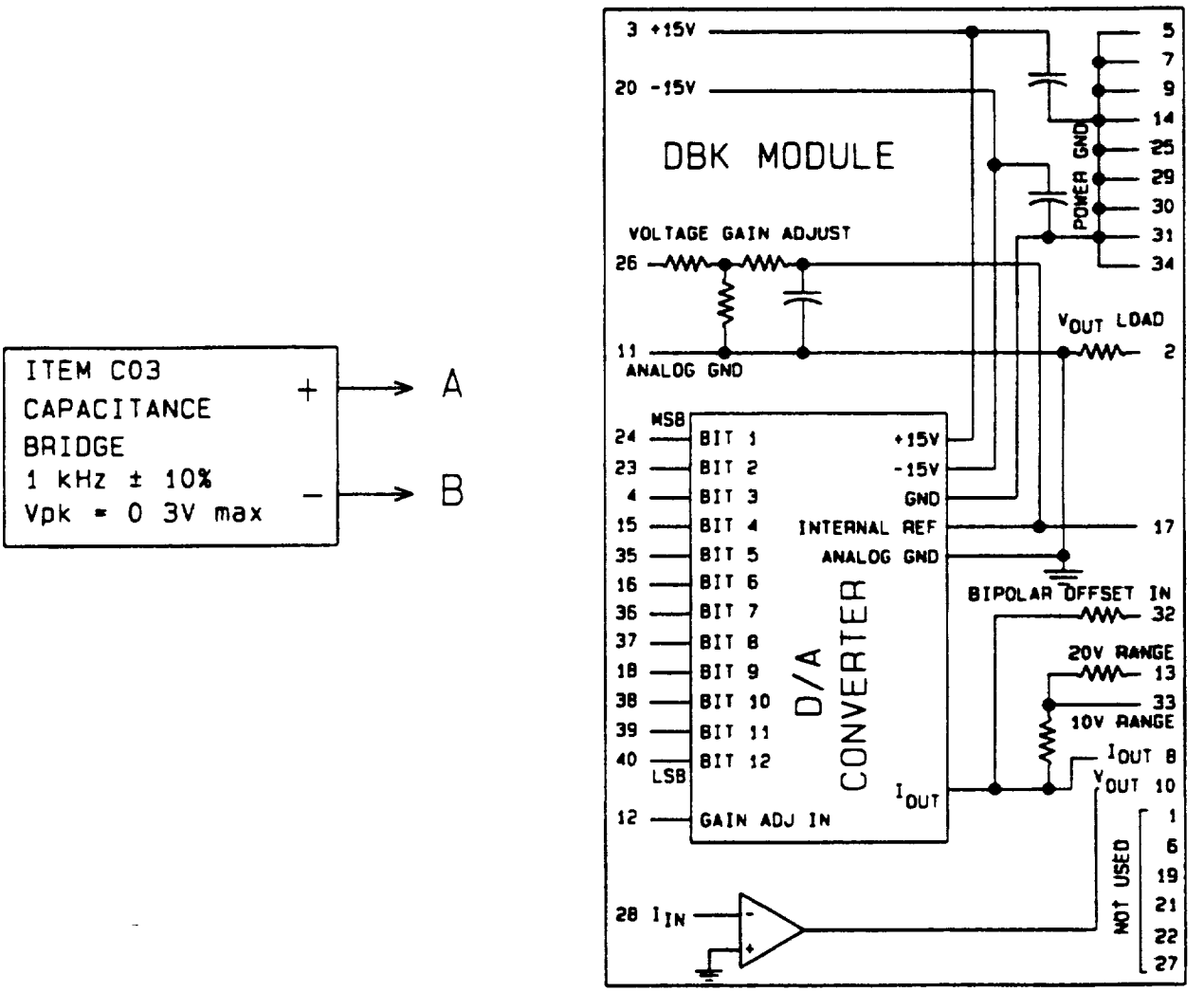
* FIGURE 10. Continuity and resistance test circuit for table VI.

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TABLE VII. Capacitance test.

Test cond	Parameter	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig.
		A to pin	B to pin	Min	Max		
F1	15 V power supply	3	30	0.75	1.25	μF	11
F2	-15 V power supply	30	20	0.75	1.25	μF	11

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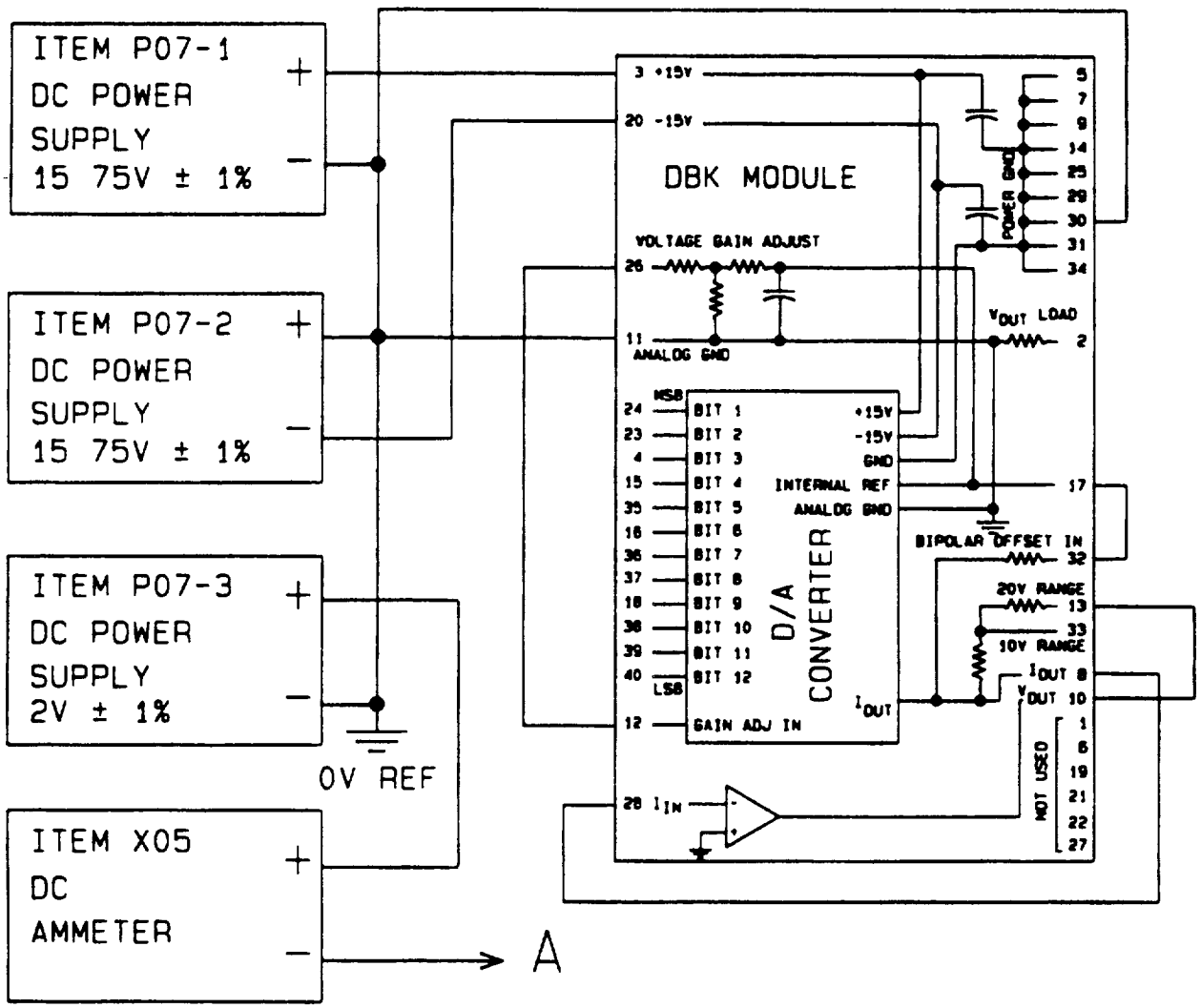
* FIGURE 11. Capacitance test circuit for table VII.

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* TABLE VIII. Digital input current (high level) test.

Test cond	Parameter	Connect A to pin	Initial limits 25°C ± 5°C ambient Min	Initial limits 25°C ± 5°C ambient Max	Unit	Test figure
G1	Bit 3	4	-10	40	μA	12
G2	Bit 4	15				
G3	Bit 6	16				
G4	Bit 9	18				
G5	Bit 2	23				
G6	Bit 1	24				
G7	Bit 5	35				
G8	Bit 7	36				
G9	Bit 8	37				
G10	Bit 10	38				
G11	Bit 11	39				
G12	Bit 12	40	-10	40	μA	12

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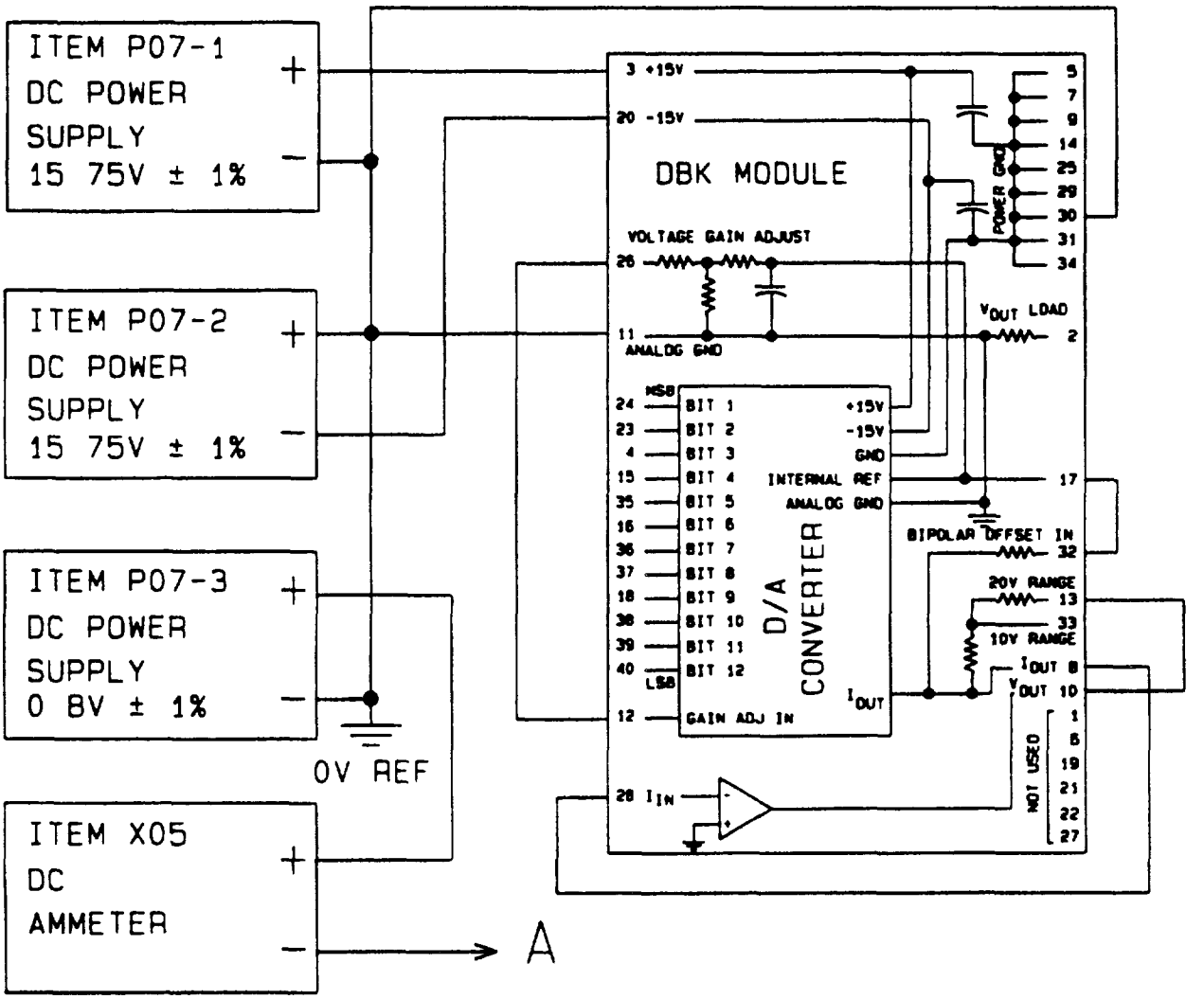
* FIGURE 12. Digital input current (high level) test circuit for table VIII.

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TABLE IX. Digital input current (low level) test.

Test cond	Parameter	Connect A to pin	Initial limits 25°C ± 5°C ambient Min	Initial limits 25°C ± 5°C ambient Max	Unit	Test figure
H1	Bit 3	4	-1.6	0.0	mA	13
H2	Bit 4	15				
H3	Bit 6	16				
H4	Bit 9	18				
H5	Bit 2	23				
H6	Bit 1	24				
H7	Bit 5	35				
H8	Bit 7	36				
H9	Bit 8	37				
H10	Bit 10	38				
H11	Bit 11	39				
H12	Bit 12	40	-1.6	0.0	mA	13

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* FIGURE 13. Digital input current (low level) test circuit for table IX.

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* TABLE X Bipolar accuracy test

Test cond	Parameter	Connect		Set		Initial limits 25°C ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	SW1	SW2	Min	Max		
J1	± 2 5 V range, positive full scale	8	10	1	1	2 4878	2 5122	V	14
J2	± 2 5 V range, zero offset	8	10	2	1	-20	20	mV	14
J3	± 2 5 V range, negative full scale	8	10	2	2	-2 5122	-2 4878	V	14
J4	± 5 V range, positive full scale	Open	10	1	1	4.9756	5.0244	V	14
J5	± 5 V range, zero offset	Open	10	2	1	-30	30	mV	14
J6	± 5 V range, negative full scale	Open	10	2	2	-5 0244	-4.9756	V	14
J7	± 10 V range, positive full scale	10	Open	1	1	9 9512	10 0488	V	14
J8	± 10 V range, zero offset	10	Open	2	1	-50	50	mV	14
J9	± 10 V range, negative full scale	10	Open	2	2	-10 0488	-9 9512	V	14

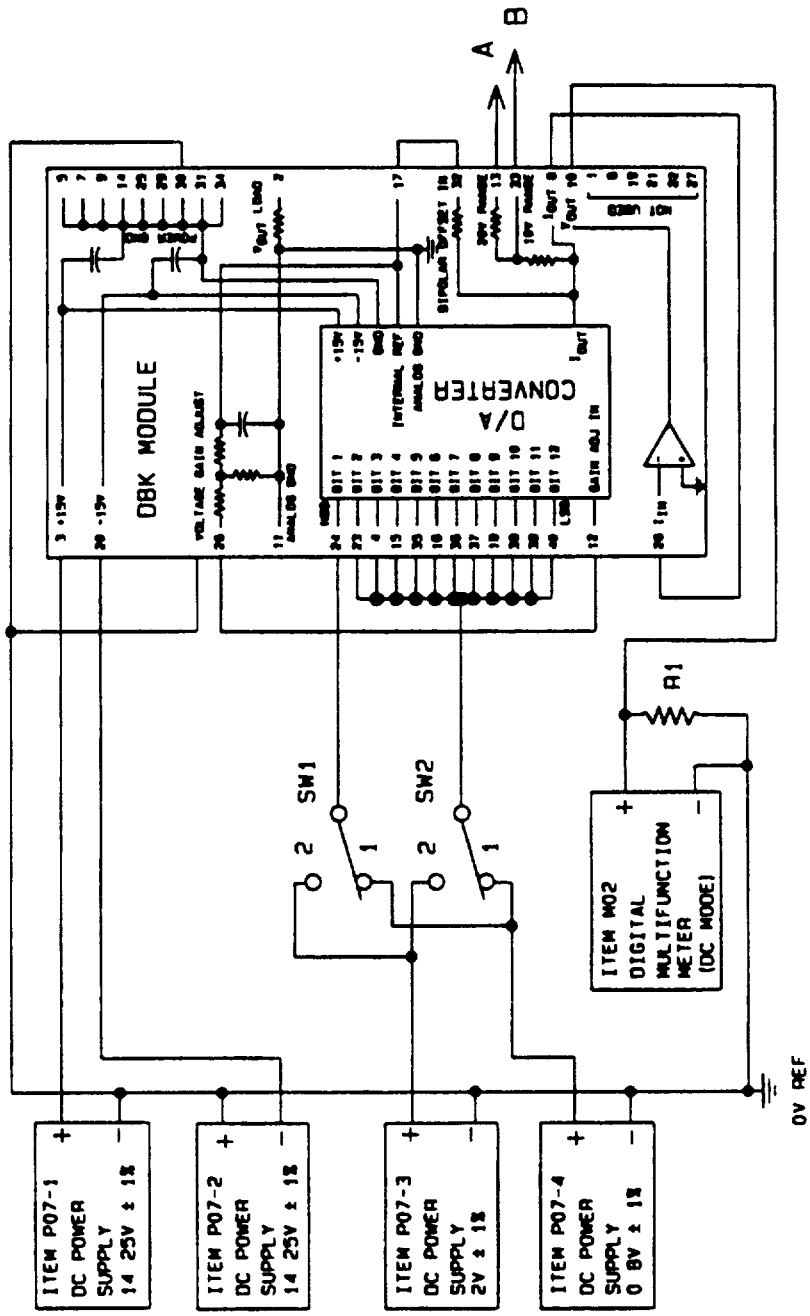
1 For brevity, table I may list bipolar accuracy in terms of LSBs where:

$$LSB = \frac{(\text{Value from table X} - \text{Full scale}) \times 2048}{\text{Full scale}}$$

For example, J1 (Min) is listed in table I as

$$\frac{(2.4878 - 2.5) \times 2048}{2\ 5} = -10 \text{ LSB}$$

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NOTES:

1. R1 = 2 kOhms ± 1 percent, 1/10 W.
2. SW1 = SW2 = SPDT switch.
3. The module shall have an initial warm up period of at least 5 minutes.

* FIGURE 14. Bipolar accuracy test circuit for table X.

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* TABLE XI. Unipolar accuracy test. 1/

Test cond	Digital input	N	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig
			A to pin	B to pin	Min	Max		
* K1	FFFh	4095	4,15,16,18, 23,24,35,36, 37,38,39,40	Open	-10.0000	-9.9951	V	15
K2	F00h	3840	4,15,23,24	16,18,35,36, 37,38,39,40	-9.3774	-9.3725		
K3	EFFh	3839	4,16,18,23, 24,35,36,37, 38,39,40	15	-9.3750	-9.3701		
K4	E00h	3584	4,23,24	15,16,18,35, 36,37,38,39, 40	-8.7524	-8.7475		
K5	DFFh	3583	15,16,18,23, 24,35,36,37, 38,39,40	4	-8.7500	-8.7451		
K6	D00h	3328	15,23,24	4,16,18,35, 36,37,38,39, 40	-8.1274	-8.1225		
K7	CFFh	3327	16,18,23,24, 35,36,37,38, 39,40	4,15	-8.1250	-8.1201	V	15

See footnote at end of table

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* TABLE XI Unipolar accuracy test - Continued 1/

Test cond	Digital input	N	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig
			A to pin	B to pin	Min	Max		
K8	C00h	3072	23,24	4,15,16,18, 35,36,37,38, 39,40	-7 5024	-7 4975	V	15
K9	BFFh	3071	4,15,16,18, 24,35,36,37, 38,39,40	23	-7 5000	-7 4951		
K10	B00h	2816	4,15,24	16,18,23,35, 36,37,38,39, 40	-6.8774	-6 8725		
K11	AFFh	2815	4,16,18,24, 35,36,37,38, 39,40	15,23	-6.8750	-6 8701		
K12	A00h	2560	4,24	15,16,18, 23,35,36,37, 38,39,40	-6 2524	-6 2475		
K13	9FFh	2559	15,16,18,24, 35,36,37,38, 39,40	4,23	-6 2500	-6 2451		
K14	900h	2304	15,24	4,16,18,23, 35,36,37,38, 39,40	-5.6274	-5.6225	V	15

See footnote at end of table

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* TABLE XI. Unipolar accuracy test - Continued 1/

Test cond	Digital input	N	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig
			A to pin	B to pin	Min	Max		
K15	8FFh	2303	16,18,24,35, 36,37,38,39, 40	4,15,23	-5 6250	-5 6201	V	15
K16	800h	2048	24	4,15,16,18, 23,35,36,37, 38,39,40	-5.0024	-4.9975		
K17	7FFh	2047	4,15,16,18, 23,35,36,37, 38,39,40	24	-5 0000	-4.9951		
K18	700h	1792	4,15,23	16,18,24,35, 36,37,38,39, 40	-4.3774	-4.3725		
K19	6FFh	1791	4,16,18,23, 35,36,37,38, 39,40	15,24	-4.3750	-4.3701		
K20	600h	1536	4,23	15,16,18,24, 35,36,37,38, 39,40	-3.7524	-3.7475		
K21	5FFh	1535	15,16,18,23, 35,36,37,38, 39,40	4,24	-3.7500	-3.7451	V	15

See footnote at end of table

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* TABLE XI Unipolar accuracy test - Continued 1/

Test cond	Digital input	N	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig
			A to pin	B to pin	Min	Max		
K22	500h	1280	15,23	4,16,18,24, 35,36,37,38, 39,40	-3 1274	-3.1225	V	15
K23	4FFh	1279	16,18,23,35, 36,37,38,39, 40	4,15,24	-3 1250	-3.1201		
K24	400h	1024	23	4,15,16,18, 24,35,36,37, 38,39,40	-2 5024	-2 4975		
K25	3FFh	1023	4,15,16,18, 35,36,37,38, 39,40	23,24	-2 5000	-2.4951		
K26	300h	768	4,15	16,18,23,24, 35,36,37,38, 39,40	-1.8774	-1.8725		
K27	2FFh	767	4,16,18,35, 36,37,38,39, 40	15,23,24	-1 8750	-1.8701		
K28	200h	512	4	15,16,18,23, 24,35,36,37, 38,39,40	-1 2524	-1.2475	V	15

See footnote at end of table

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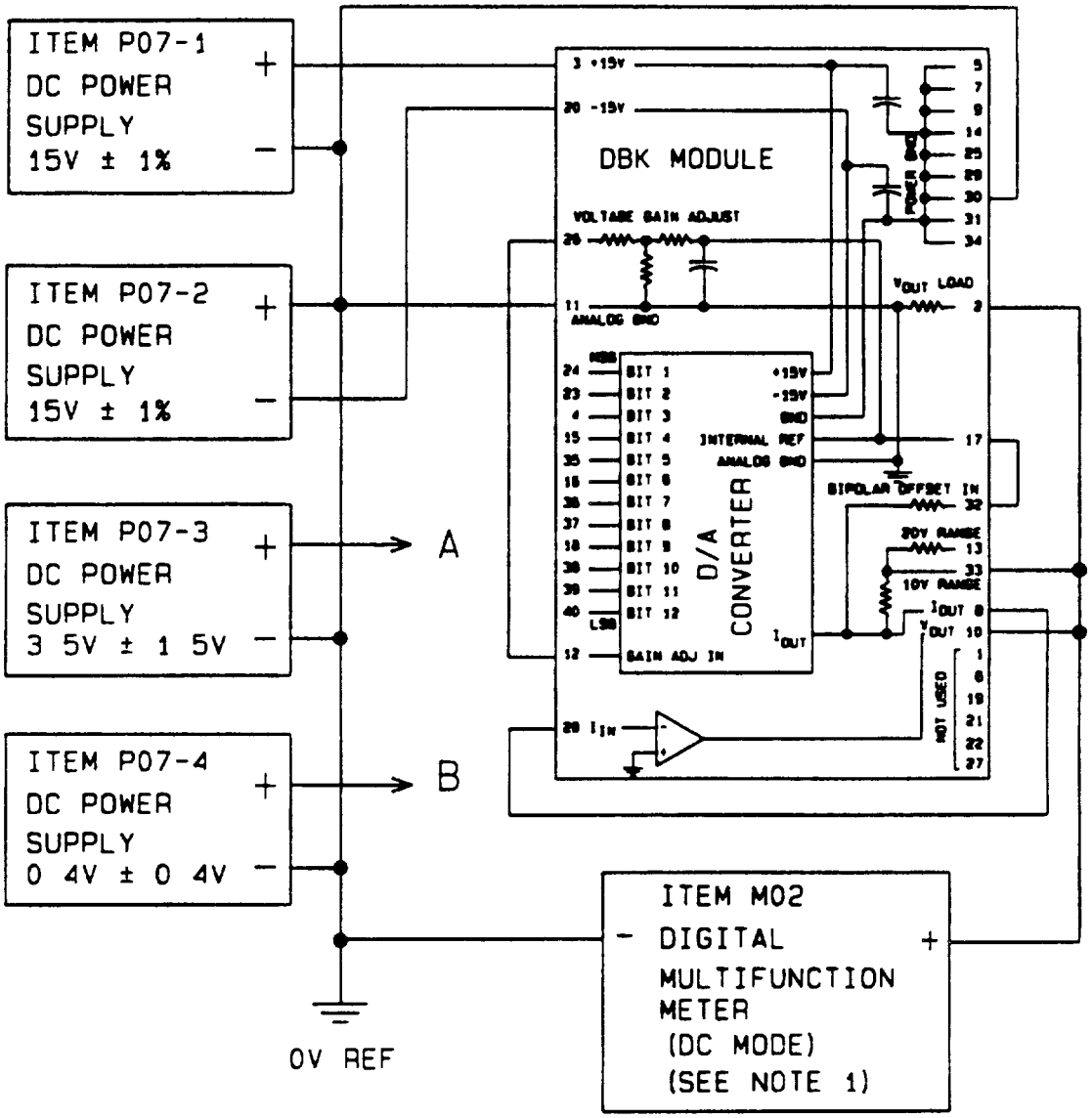
* TABLE XI Unipolar accuracy test - Continued. 1/

Test cond	Digital input	N	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig.
			A to pin	B to pin	Min	Max		
K29	1FFh	511	15,16,18,35, 36,37,38,39, 40	4,23,24	-1.2500	-1 2451	V	15
K30	100h	256	15	4,16,18,23, 24,35,36,37, 38,39,40	-0.6274	-0.6225		
K31	0FFh	255	16,18,35,36, 37,38,39,40	4,15,23,24	-0.6250	-0.6201		
K32	000h	0	Open	4,15,16,18, 23,24,35,36, 37,38,39,40	-0.0024	0.0024	V	15

1/ For brevity, unipolar accuracy is listed in table I in terms of LSBs, where:

LSB = (|Full scale|) / 4096 = (| -10.0000 |) / 4096 = 2.44 mV

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NOTES:

1. Item M02 must have an accuracy of ± (0.006 percent of reading + 0.001 percent of range) or better on the 10 V full scale range. Not all equipment listed in MIL-STD-1665 for item M02 will meet this requirement.

2. All measurements for this test shall be made without interruption to eliminate errors due to test equipment and module drift. The module shall have an initial warm up period of at least 5 minutes.

* FIGURE 15. Unipolar accuracy test circuit for table XI.

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* TABLE XII. Differential linearity error test. 1/

Test cond	Calculations 2/	Initial limits 25°C ± 5°C ambient		Unit
		Min	Max	
L1	K3 - K2	0	4.88	mV
L2	K5 - K4			
L3	K7 - K6			
L4	K9 - K8			
L5	K11 - K10			
L6	K13 - K12			
L7	K15 - K14			
L8	K17 - K16			
L9	K19 - K18			
L10	K21 - K20			
L11	K23 - K22			
L12	K25 - K24			
L13	K27 - K26			
L14	K29 - K28			
L15	K31 - K30	0	4.88	mV

1/ This test guarantees monotonicity only for the codes tested.

2/ Test conditions L1 through L15 are calculations based on readings taken from test conditions K1 through K31 as defined in table XI. The results of subtracting the readings taken from the specified test conditions shall meet the limits shown.

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* TABLE XIII. Linearity error test.

Test cond	X 1/	N	Initial limits 25°C ± 5°C ambient		Unit
			Min	Max	
M1	K2	3840	-2.44	2.44	mV
M2	K3	3839			
M3	K4	3584			
M4	K5	3583			
M5	K6	3328			
M6	K7	3327			
M7	K8	3072			
M8	K9	3071			
M9	K10	2816			
M10	K11	2815			
M11	K12	2560			
M12	K13	2559			
M13	K14	2304			
M14	K15	2303			
M15	K16	2048			
M16	K17	2047			
M17	K18	1792			
M18	K19	1791			
M19	K20	1536			
M20	K21	1535	-2.44	2.44	mV

See footnote at end of table

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* TABLE XIII. Linearity error test - Continued.

Test cond	X 1/	N	Initial limits 25°C ± 5°C ambient		Unit
			Min	Max	
M21	K22	1280	-2.44	2.44	mV
M22	K23	1279			
M23	K24	1024			
M24	K25	1023			
M25	K26	768			
M26	K27	767			
M27	K28	512			
M28	K29	511			
M29	K30	256			
M30	K31	255	-2.44	2.44	mV

1/ Test conditions M1 through M30 are calculations. The results of the following calculation shall meet the limits shown:

X - Offset + N x Increment where:

X is the value of the reading taken for the indicated test conditions K2 through K31 as defined in table XI.

Offset is the value of the reading taken for test condition K32 as defined in table XI.

N is taken from the above table.

Increment is (K32 - K1) / 4095.

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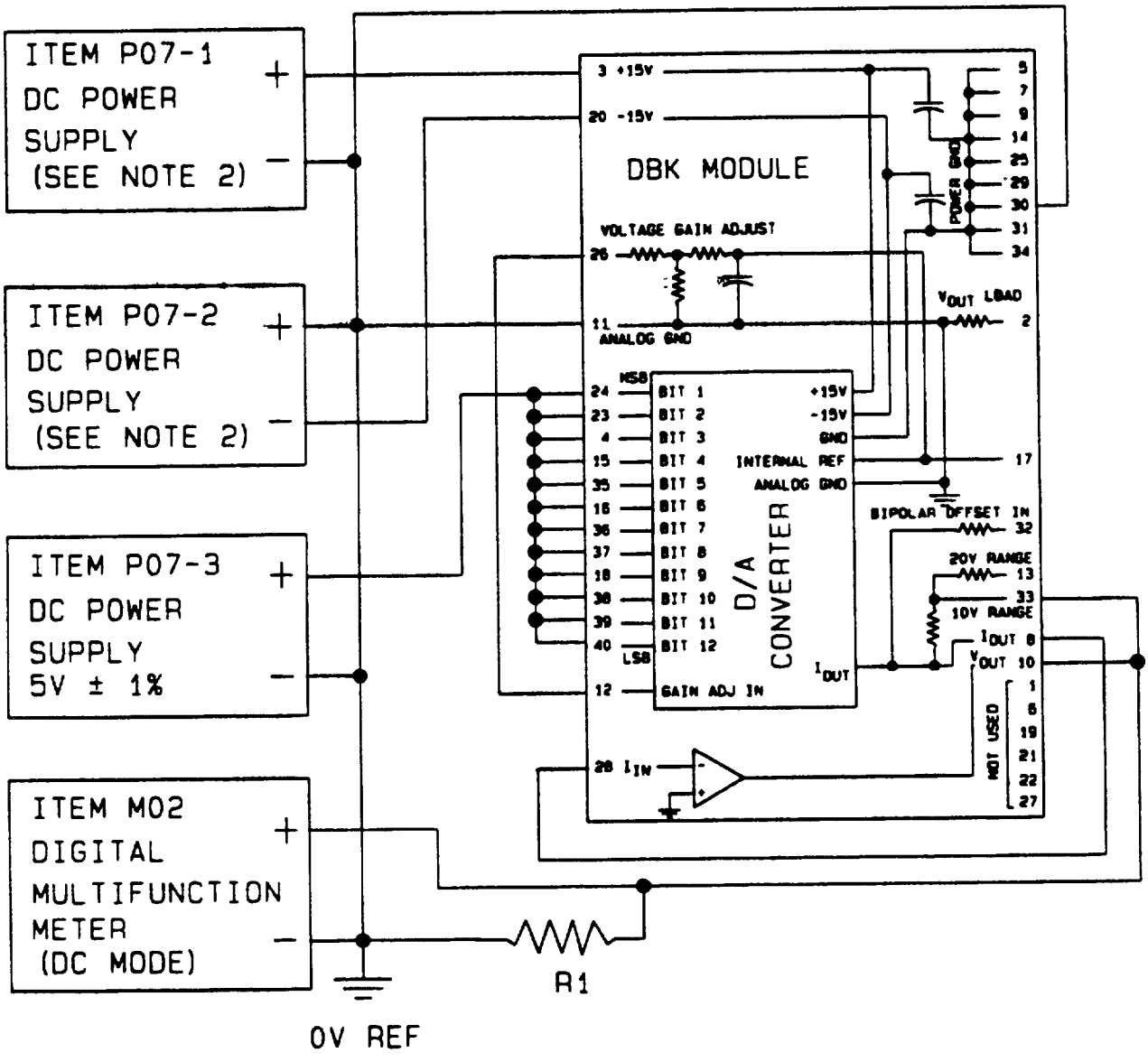
* TABLE XIV. Power supply sensitivity test. 1/

Test cond	Item changed	Initial voltage 1/	Percent change	Test voltage 1/	Initial limits 25°C ± 5°C ambient		Unit	Test fig.
					Min	Max		
N1	P07-1	15.00 V	+5	15.75 V	-6.1	6.1	mV	16
N2	P07-1	15.00 V	-5	14.25 V				
N3	P07-2	15.00 V	+5	15.75 V				
N4	P07-2	15.00 V	-5	14.25 V	-6.1	6.1	mV	16

1/ Test conditions N1 through N4 shall be performed as follows:

(a) Items P07-1 and P07-2 shall be set at 15.00 V ± 2 percent.
(b) Record the reading on item M02 to at least 0.1 mV resolution.
(c) Adjust the item changed to the specified test voltage ± 2 percent.
(d) Record the reading on item M02 to at least 0.1 mV resolution.
(e) Subtract the reading taken in step (d) from that taken in step (b). The result shall meet the limits shown.

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NOTES:

1. R1 = 2 kOhms ± 1 percent, 1/10 W.
2. See footnote to table XIV for voltage settings.

* FIGURE 16. Power supply sensitivity test circuit for table XIV.

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* TABLE XV. Settling time test.

Test cond	Initial voltage	Final voltage	D in mV	Set SW1 to	Connect B to pin	Initial limits 25°C ± 5°C ambient		Unit	Test figure
						Min	Max		
P1	-10	10	2.5	2	23	0.0	5.0	µS	17,18
P2	-10	10	10	2			3.0		
P3	10	-10	2.5	1			5.0		
P4	10	-10	10		23		3.0		
P5	0	-10	2.5		A		4.5		
P6	0	-10	10	1			2.5		
P7	-10	0	2.5	Off			4.5		
P8	-10	0	10	Off	A	0.0	2.5	µS	17,18

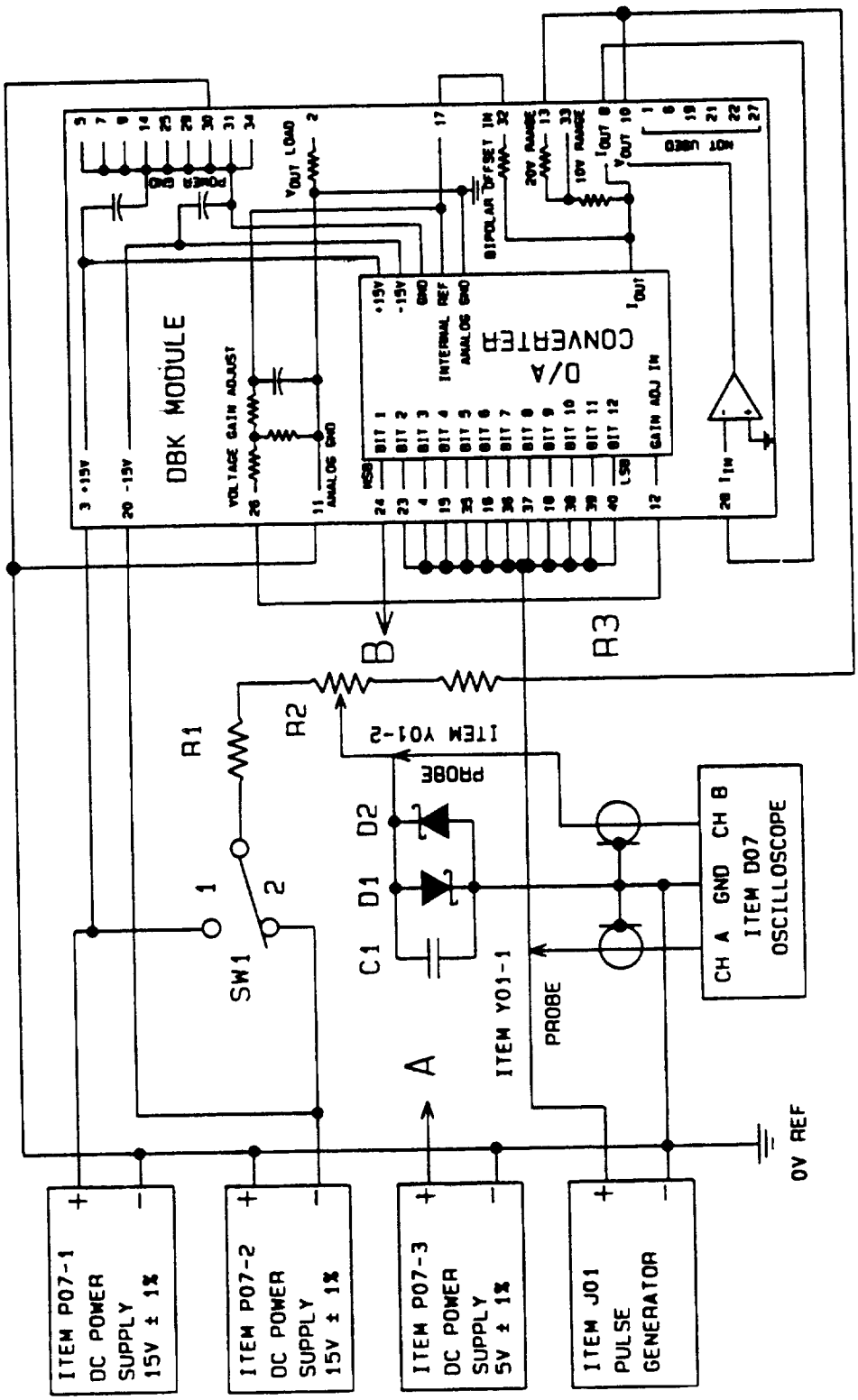
NOTES:

1. Test conditions P1 and P2 shall be performed as follows:
   (a) Make connections as specified.
   (b) For each test condition, adjust R2 until the high level of the pulse on channel B of item D07 is 0 ± 1 mV.
   (c) Measure tSLH.

2. Test conditions P3 through P6 shall be performed as follows:
   (a) Make connections as specified.
   (b) For each test condition, adjust R2 until the low level of the pulse on channel B of item D07 is -10 V ± 1 mV.
   (c) Measure tSHL.

3. Test conditions P7 and P8 shall be performed as follows:
   (a) Make connections as specified.
   (b) Measure tSLH.

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* FIGURE 17. Settling time test circuit for table XV.

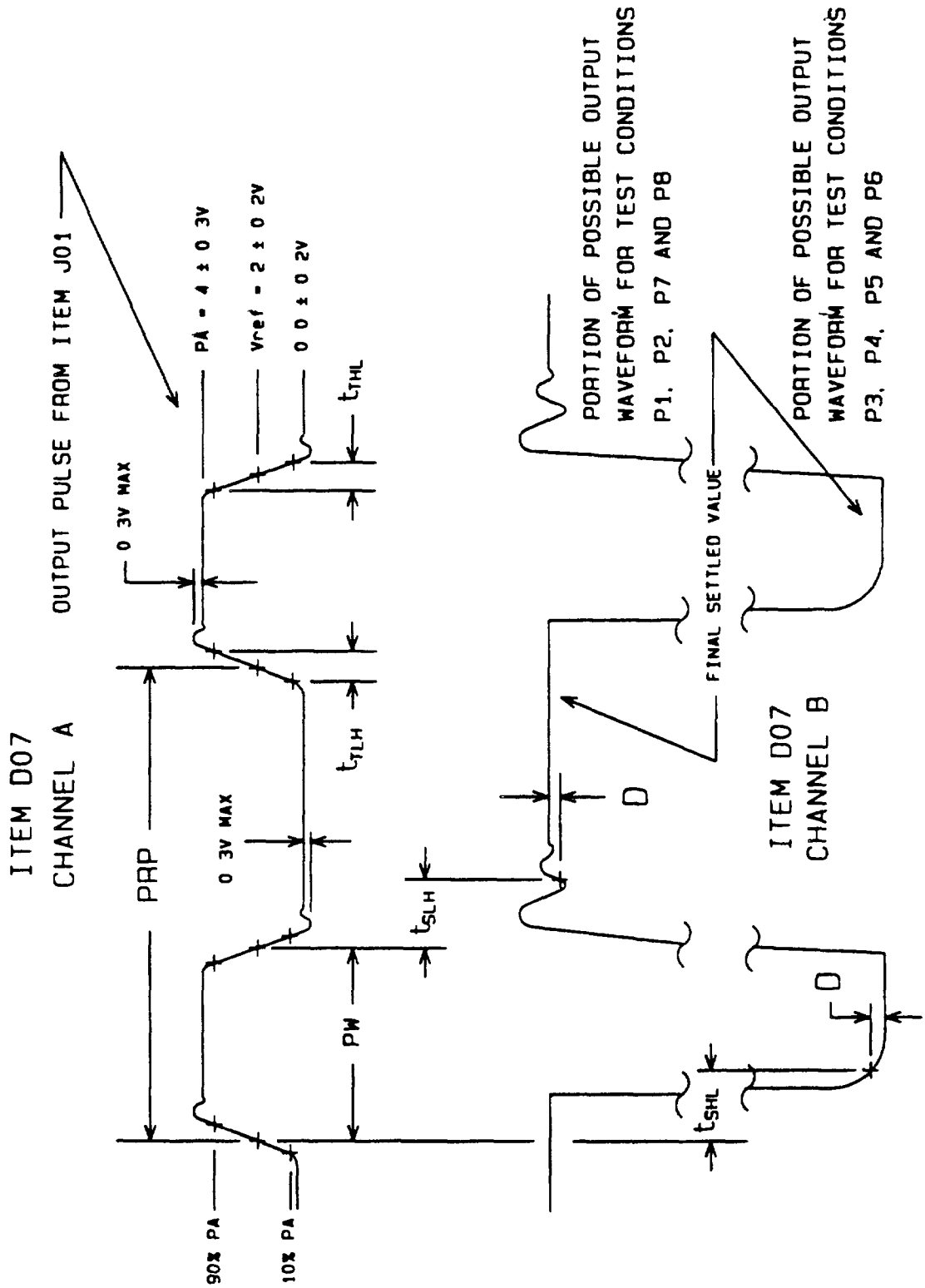
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NOTES:

1. R1 = 3.16 kOhms ± 1 percent, 1/8 W.
2. R2 = 500 Ohms ± 5 percent, 1/4 W, multiturn potentiometer
3. R3 = 2 kOhms ± 1 percent, 1/8 W.
4. D1 = D2 = Schottky diodes.
5. SW1 = SPDT switch (center off).
6. C1 = 50 pF ± 10 percent (including probe capacitance).

* FIGURE 17. Settling time test circuit for table XV.

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* FIGURE 18. Settling time test circuit waveforms for table XV.

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NOTES:

1. Item J01 pulse parameters:
   PRP = 100 ± 1 μS.
   PW = 50 ± 0.5 μS.
   tTLH = tTHL = 20 ± 10 nS.

2. Obtain values for D from table XV.

* FIGURE 18. Settling time test circuit waveforms for table XV - Continued.

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